

Multinomial Discrete Choice: IIA Logit

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Grad IO

Most decisions agents make are not necessarily binary:

- ▶ Choosing a level of schooling (or a major).
- ▶ Choosing an occupation.
- ▶ Choosing a partner.
- ▶ Choosing where to live.
- ▶ Choosing a brand of (yogurt, laundry detergent, orange juice, cars, etc.).

Nonparametric Setup

We consider a **multinomial discrete choice**:

- ▶ in period t
- ▶ with $\mathcal{J}_t$ alternatives.
- ▶ subscript individual agents by i .
- ▶ agents choose $j \in \mathcal{J}_t$ with probability σ_{ijt} .
- ▶ Agent i receives utility U_{ijt} for choosing j .
- ▶ Choice is exhaustive and mutually exclusive.

Consider the simple example ($t = 1$):

$$\sigma_{ij} = \mathbb{P}(U_{ij} > U_{ik} \quad \forall k \neq j)$$

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Now consider separating the utility into the **observed** V_{ij} and **unobserved** components ε_{ij} .

$$\begin{aligned}\sigma_{ij} &= \mathbb{P}(U_{ij} > U_{ik} \quad \forall k \neq j) \\ &= \mathbb{P}(V_{ij} + \varepsilon_{ij} > V_{ik} + \varepsilon_{ik} \quad \forall k \neq j) \\ &= \mathbb{P}(\varepsilon_{ij} - \varepsilon_{ik} > V_{ik} - V_{ij} \quad \forall k \neq j)\end{aligned}$$

It is helpful to define $f(\varepsilon_{\mathbf{i}})$ as the J vector of individual i 's unobserved utility.

$$\begin{aligned}\sigma_{ij} &= \mathbb{P}(\varepsilon_{ij} - \varepsilon_{ik} > V_{ik} - V_{ij} \quad \forall k \neq j) \\ &= \int I(\varepsilon_{ij} - \varepsilon_{ik} > V_{ik} - V_{ij}) f(\varepsilon_{\mathbf{i}}) d\varepsilon_{\mathbf{i}}\end{aligned}$$

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In order to compute the choice probabilities, we must perform a J dimensional integral over $f(\boldsymbol{\varepsilon}_{\mathbf{i}})$.

$$\sigma_{ij} = \int I(\varepsilon_{ij} - \varepsilon_{ik} > V_{ik} - V_{ij}) f(\boldsymbol{\varepsilon}_{\mathbf{i}}) d\boldsymbol{\varepsilon}_{\mathbf{i}}$$

There are some choices that make our life easier

- ▶ Multivariate normal: $\boldsymbol{\varepsilon}_{\mathbf{i}} \sim N(0, \Omega)$. $\rightarrow$ **multinomial probit**.
- ▶ Gumbel/Type 1 EV: $f(\boldsymbol{\varepsilon}_{\mathbf{i}}) = e^{-\varepsilon_{ij}} e^{-e^{-\varepsilon_{ij}}}$ and $F(\boldsymbol{\varepsilon}_{\mathbf{i}}) = 1 - e^{-e^{-\varepsilon_{ij}}}$ $\rightarrow$ **multinomial logit**
- ▶ There are also heteroskedastic variants of the Type I EV/ Logit framework.

Allowing for a continuous density with full support $(-\infty, \infty)$ errors provide two key features:

- ▶ Smoothness: σ_{ij} is everywhere continuously differentiable in V_{ij} .
- ▶ Bound $\sigma_{ij} \in (0, 1)$ so that we can rationalize any observed pattern in the data.
 - Caveat: zero and one (interpretation).
- ▶ What does ε_{ij} really mean? (unobserved utility, idiosyncratic tastes, etc.)

Basic Identification

- ▶ Only differences in utility matter: $\mathbb{P}(\varepsilon_{ij} - \varepsilon_{ik} > V_{ik} - V_{ij} \quad \forall k \neq j)$
- ▶ Adding constants is irrelevant: if $U_{ij} > U_{ik}$ then $U_{ij} + a > U_{ik} + a$.
- ▶ Only differences in alternative specific constants can be identified

$$U_b = v_{ib} + k_b + \varepsilon_{ib}$$

$$U_c = v_{ic} + k_c + \varepsilon_{ic}$$

only $d = k_b - k_c$ is identified.

- ▶ This means that we can only include $J - 1$ such k 's and need to normalize one to zero. (Much like fixed effects).
- ▶ We cannot have individual specific factors that enter the utility of all options such as income θY_i . We can allow for interactions between individual and choice characteristics $\theta p_j / Y_i$.

$$U_b = v_b + \theta y_i + \varepsilon_b$$

$$U_c = v_c + \theta y_i + \varepsilon_c$$

- ▶ Technically we can't really fully specify $f(\varepsilon_{\mathbf{i}})$ since we can always re-normalize:
 $\widetilde{\varepsilon}_{ijk} = \varepsilon_{ij} - \varepsilon_{ik}$ and write $g(\widetilde{\varepsilon}_{\mathbf{ik}})$. Thus any $g(\widetilde{\varepsilon}_{\mathbf{ik}})$ is consistent with infinitely many $f(\varepsilon_{\mathbf{i}})$.
- ▶ Logit pins down $f(\varepsilon_{\mathbf{i}})$ sufficiently with parametric restrictions.
- ▶ Probit does not. We must generally normalize one dimension of $f(\varepsilon_{\mathbf{i}})$ in the probit model. Usually a diagonal term of Ω so that $\omega_{11} = 1$ for example. (Actually we need to do more!).

Basic Identification: Scale

- ▶ Consider: $U_{ij}^0 = V_{ij} + \varepsilon_{ij}$ and $U_{ij}^1 = \lambda V_{ij} + \lambda \varepsilon_{ij}$ with $\lambda > 0$. Multiplying by constant λ factor doesn't change any statements about $U_{ij} > U_{ik}$.
- ▶ We normalize this by fixing the variance of ε_{ij} since $Var(\lambda \varepsilon_{ij}) = a^2 \lambda^2$.
- ▶ Normalizing this variance normalizes the scale of utility.
- ▶ For the logit case the variance is normalized to $\pi^2/6$. (this emerges as a constant of integration to guarantee a proper density).

Observed Heteroskedasticity

Consider the case where $Var(\varepsilon_{ib}) = a^2$ and $Var(\varepsilon_{ic}) = k^2 a^2$:

- We can estimate

$$U_{ib} = v_{ib} + \varepsilon_{ib}$$

$$U_{ic} = v_{ic} + \varepsilon_{ic}$$

becomes:

$$U_{ib} = v_{ib} + \varepsilon_{ib}$$

$$U_{ic} = v_{ic}/k + \varepsilon_{ic}$$

- Some interpret this as saying that in segment C the unobserved factors are $\hat{k}$ times larger.

Different ways to look at identification

- ▶ Are we interested in non-parametric identification of V_{ij} , specifying $f(\epsilon_i)$?
- ▶ Or are we interested in non-parametric identification of U_{ij} . (Generally hard).
 - Generally we require a large support (special-regressor) or “completeness” condition.
 - Lewbel (2000) does random utility with additively separable but nonparametric error.
 - Berry and Haile (2015) with non-separable error (and endogeneity).

- ▶ Multinomial Logit (Gumbel/Type I EV) has closed form choice probabilities

$$\sigma_{ij} = \frac{e^{V_{ij}}}{\sum_k e^{V_{ik}}}$$

- ▶ Often we approximate $V_{ij} \approx X_{ik}\beta$ with something linear in parameters.

Expected maximum also has closed form:

$$\mathbb{E}[\max_j U_{ij}] = \log \left(\sum_j \exp[V_{ij}] \right) + C$$

Logit Inclusive Value is helpful for several reasons

- ▶ Expected utility of best option (without knowledge of ε_i) does not depend on ε_{ij} .
- ▶ This is a globally **convex** function in V_{ij} (more on that in a moment).
- ▶ Allows simple computation of ΔCS for consumer welfare (but not CS itself).

The Surplus Function: Williams-Daly-Zachary

Call the expected maximum the **social surplus function**:

$$G(V_{i1}, \dots, V_{iJ}) \equiv \mathbb{E}[\max_j V_{ij} + \varepsilon_{ij}]$$

For **any** additive random utility model (not just logit), the **Williams-Daly-Zachary theorem** says G is convex with:

$$\frac{\partial G}{\partial V_{ij}}(\mathbf{V}_i) = \sigma_{ij}(\mathbf{V}_i)$$

- ▶ The gradient of the surplus function is the vector of choice probabilities: the discrete-choice analogue of Roy's Identity / Shephard's Lemma.
- ▶ Logit check: $\nabla_j \log \left(\sum_k \exp[V_{ik}] \right) = \frac{e^{V_{ij}}}{\sum_k e^{V_{ik}}} = \sigma_{ij}$.
- ▶ The gradient of smooth-max is softmax — the next slides are secretly this theorem.
- ▶ Later: Hotz-Miller **invert** this mapping (probabilities $\rightarrow$ values) for dynamic discrete choice; for a convex-duality treatment see Chiong, Galichon, Shum (2016).

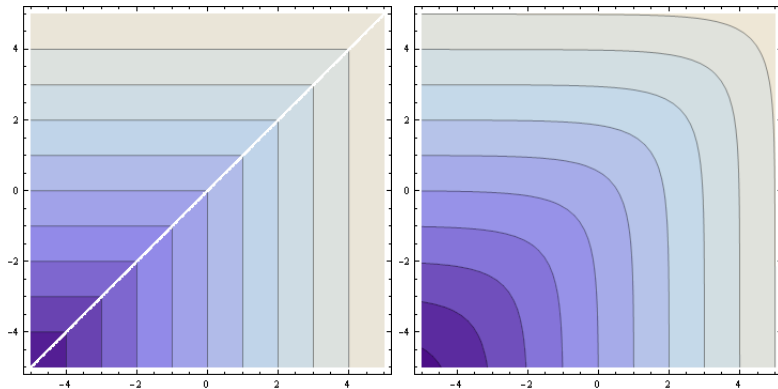
Multinomial Logit goes by a lot of names in various literatures

- ▶ The problem of multiple choice is often called **multiclass classification** or **softmax regression** in other literatures.
- ▶ In general these models assume you have individual level data

Statistics/Computer Science offer an alternative interpretation

- ▶ Sometimes this is called **softmax** regression.
- ▶ Think of this as a continuous/concave approximation to the maximum.
- ▶ Consider $\max\{x, y\}$ vs $\log(\exp(x) + \exp(y))$. The exp exaggerates the differences between x and y so that the larger term dominates.
- ▶ We can accomplish this by rescaling k : $\log(\exp(kx) + \exp(ky))/k$ as k becomes large the derivatives become infinite and this approximates the “hard” maximum.
- ▶ $g(1, 2) = 2.31$, but $g(10, 20) = 20.00004$.

Alternative Interpretation



What is actually identified here?

- ▶ Helpful to look at the ratio of two choice probabilities

$$\frac{\sigma_{ij}(\theta)}{\sigma_{ik}(\theta)} = \frac{e^{V_{ij}}}{e^{V_{ik}}} = e^{V_{ij}-V_{ik}}$$

- ▶ We only identify the **difference in indirect utilities** not the levels.
- ▶ The ratio of choice probabilities for j and k depends only on j and k and not on any alternative l , this is known as **independence of irrelevant alternatives**.
- ▶ For some (Luce (1959)) IIA was an attractive property for axiomatizing choice. (A feature or a bug?)
- ▶ In fact the logit was derived in the search for a statistical model that satisfied various axioms.

As another idea suppose we add a constant C to each β_j .

$$\sigma_{ij} = \frac{\exp[\mathbf{x}_i(\beta_j + C)]}{\sum_k \exp[\mathbf{x}_i(\beta_k + C)]} = \frac{\exp[\mathbf{x}_i C] \exp[\mathbf{x}_i \beta_j]}{\exp[\mathbf{x}_i C] \sum_k \exp[\mathbf{x}_i \beta_k]}$$

This has no effect. That means we need to fix a normalization C .

The most convenient is generally that $C = -\beta_K$.

- ▶ We normalize one of the choices to provide a utility of zero.
- ▶ We actually already made another normalization. Does anyone know which?

The most sensible normalization in demand settings is to allow for an **outside option** which produces no utility in expectation so that $e^{V_{i0}} = e^0 = 1$:

$$\sigma_{ij} = \frac{e^{V_{ij}}}{1 + \sum_k e^{V_{ik}}}$$

- ▶ Hopefully the choice of outside option is well defined: not buying a yogurt, buying some other used car, etc.
- ▶ Now this resembles the binomial logit model more closely.

- ▶ For the logit, $U_{ij}^* = V_{ij} + \varepsilon_{ij}^*$ has $Var(\varepsilon^*) = a^2\pi^2/6$; dividing by the scale parameter a gives $U_{ij} = V_{ij}/a + \varepsilon_{ij}$ and $Var(\varepsilon) = \pi^2/6$

$$\sigma_{ij} = \frac{e^{V_{ij}/a}}{\sum_k e^{V_{ik}/a}} \approx \frac{e^{\beta^*/a \cdot x_{ij}}}{\sum_k e^{\beta^*/a \cdot x_{ik}}}$$

- ▶ Every coefficient β is rescaled by a . This implies that only the ratio β^*/a is identified.
- ▶ Coefficients are relative to variance of unobserved factors. More unobserved variance $\rightarrow$ smaller β .
- ▶ Ratio β_1/β_2 is invariant to the scale parameter a . (**marginal rate of substitution**).

The well known critique:

- ▶ You can choose to go to work on a car c or blue bus bb . $\sigma_c = \sigma_{bb} = \frac{1}{2}$ so that $\frac{\sigma_c}{\sigma_{bb}} = 1$.
- ▶ Now we introduce a red bus rb that is identical to bb . Then $\frac{\sigma_{rb}}{\sigma_{bb}} = 1$ and $\sigma_c = \sigma_{bb} = \sigma_{rb} = \frac{1}{3}$ as the logit model predicts.
- ▶ In reality we don't expect painting a bus red would change the number of individuals who drive a car so we would anticipate $\sigma_c = \frac{1}{2}$ and $\sigma_{bb} = \sigma_{rb} = \frac{1}{4}$.
- ▶ We may not encounter too many cases where $\rho_{\varepsilon_{ik}, \varepsilon_{ij}} \approx 1$, but we have many cases where this $\rho_{\varepsilon_{ik}, \varepsilon_{ij}} \neq 0$
- ▶ What we need is the ratio of probabilities to change when we introduce a third option!

- ▶ IIA implies that we can obtain consistent estimates for β on any subset of alternatives.
- ▶ This means instead of using all $\mathcal{J}$ alternatives in the choice set, we could estimate on some subset $\mathcal{S} \subset \mathcal{J}$.
- ▶ This used to be a way to reduce the computational burden of estimation (not clear this is an issue in 21st century).
- ▶ Sometimes we have **choice based samples** where we oversample people who choose a particular alternative. Manski and Lerman (1977) show we can get consistent estimates for all but the ASC. This requires knowledge of the difference between the true (population) rate A_j and the choice-based sample rate H_j .
- ▶ Hausman proposes a specification test of the logit model: estimate on the full dataset to get $\hat{\beta}$, construct a smaller subsample $\mathcal{S}^k \subset \mathcal{J}$ and $\hat{\beta}^k$ for one or more subsets k . If $|\hat{\beta}^k - \hat{\beta}|$ is small enough.

Own and Cross Elasticity

For the linear V_{ij} case we have that $\frac{\partial V_{ij}}{\partial z_{ij}} = \beta_z$.

$$\frac{\partial \sigma_{ij}}{\partial z_{ij}} = \sigma_{ij}(1 - \sigma_{ij}) \frac{\partial V_{ij}}{\partial z_{ij}}$$

And Elasticity: $\frac{\partial \log \sigma_{ij}}{\partial \log z_{ij}} = \sigma_{ij}(1 - \sigma_{ij}) \frac{\partial V_{ij}}{\partial z_{ij}} \frac{z_{ij}}{\sigma_{ij}} = (1 - \sigma_{ij}) z_{ij} \frac{\partial V_{ij}}{\partial z_{ij}}$

With cross effects: $\frac{\partial \sigma_{ij}}{\partial z_{ik}} = -\sigma_{ij}\sigma_{ik} \frac{\partial V_{ik}}{\partial z_{ik}}$

and elasticity: $\frac{\partial \log \sigma_{ij}}{\partial \log z_{ik}} = -\sigma_{ik} z_{ik} \frac{\partial V_{ik}}{\partial z_{ik}}$

- This implies that $\eta_{jj} = \frac{\partial \sigma_{ij}}{\partial p_j} \frac{p_j}{\sigma_{ij}} = \beta_p \cdot p_j \cdot (1 - \sigma_{ij})$.
- The price elasticity is increasing in own price! (Why is this a bad idea?)
- Also mechanical relationship between elasticity and **share** so that popular products necessarily have higher markups (holding fixed prices).

Proportional Substitution and Diversion

Cross elasticity doesn't really depend on j :

$$\frac{\partial \log \sigma_{ij}}{\partial \log z_{ik}} = -\sigma_{ik} z_{ik} \underbrace{\frac{\partial V_{ik}}{\partial z_{ik}}}_{\beta_z}.$$

Recall the diversion ratio:

$$D_{jk} = \frac{\frac{\partial \sigma_{ik}}{\partial p_j}}{\left| \frac{\partial \sigma_{ij}}{\partial p_j} \right|} = \frac{\beta_p \sigma_{ik} \sigma_{ij}}{\beta_p \sigma_{ij} (1 - \sigma_{ij})} = \frac{\sigma_{ik}}{1 - \sigma_{ij}}$$

- ▶ This leads to the idea of **proportional substitution**. As option k gets better it proportionally reduces the shares of the all other choices.
- ▶ Likewise removing an option j means that $\tilde{\sigma}_{ik}(\mathcal{J} \setminus j) = \frac{\sigma_{ik}}{1 - \sigma_{ij}}$ for all other k .
- ▶ IIA/Logit means **constant diversion ratios**. This might be a desirable property but probably not.

Welfare: Consumer Surplus from the Inclusive Value

With utility linear in price, $V_{ij} = v_{ij} - \alpha p_j$, α is the marginal utility of income and (Small and Rosen 1981):

$$\mathbb{E}[CS_i] = \frac{1}{\alpha} \log \left(\sum_{k \in \mathcal{J}} \exp[V_{ik}] \right) + C$$

For any counterfactual (price change, new product, removal) the constant C differences out:

$$\Delta CS_i = \frac{1}{\alpha} \left[\log \left(\sum_{k \in \mathcal{J}^1} \exp[V_{ik}^1] \right) - \log \left(\sum_{k \in \mathcal{J}^0} \exp[V_{ik}^0] \right) \right]$$

- ▶ Constant α means **no income effects**: $CV = EV = \Delta CS$.
- ▶ With a random α_i (mixed logit) the $\frac{1}{\alpha_i}$ belongs inside the expectation over types and the clean closed form disappears (more next week).

Worked Example: The Welfare Cost of Removing a Product

Three inside goods with $V_i = (1, 0.5, 0.5)$, outside good $V_{i0} = 0$, and $\alpha = 1$ (so utils are dollars):

$$\Delta CS_i = \frac{1}{\alpha} \left[\underbrace{\log(1 + e^{0.5} + e^{0.5})}_{=4.30} - \underbrace{\log(1 + e^1 + e^{0.5} + e^{0.5})}_{=7.02} \right] = -0.49$$

- For logit this collapses to $\Delta CS_i = \frac{1}{\alpha} \log(1 - \sigma_{ij})$: the welfare cost of removing j depends **only on its share** ($\sigma_{i1} = 0.39$ here) – IIA strikes again.
- This is Hausman (1997)'s Apple-Cinnamon Cheerios question. In product space he had to find a **virtual (choke) price** p_j^* that zeroes out demand; in discrete choice we just drop j from $\mathcal{J}$ and the log-sum-exp does the rest.
- Same object as $WTP_i(j)$: reappears in HW2 and for hospital networks (Capps, Dranove, Satterthwaite 2003) later in the course.
- Caveat: full-support ε_{ij} means **every** new product mechanically adds surplus (new-goods bias: Petrin 2002; more in the micro data lecture).

CES and Logit: Two Sides of the Same Coin

Recall the CES verdict from last week: one elasticity, one markup, “hard to do IO here.” Look familiar?

	CES	Logit
Own elasticity	$-\sigma$ (same for all goods)	$-\alpha p_j(1 - \sigma_{ij}) \approx -\alpha p_j$
Substitution	proportional to expenditure share	proportional to share (IIA)
Markup	$p = mc/\rho$ (same for all goods)	$p_j - mc_j = \frac{1}{\alpha(1-\sigma_{ij})}$

- ▶ Anderson, de Palma, Thisse (1992): a CES representative consumer and a population of logit consumers generate the **same** aggregate demand system (CES is homothetic with continuous quantities; logit is unit demand — otherwise the same model).
- ▶ The fixes rhyme too: nested logit $\leftrightarrow$ nested CES; mixed logit $\leftrightarrow$ random-coefficients CES (both workhorses in Trade and Macro).
- ▶ IO speaks discrete choice because we want product-level $(p_{jt}, x_{jt}, \xi_{jt})$ — but the pathologies and their cures translate across languages.

Thanks!
